

strong duality holds

no duality gap



structural lens

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Implications

- We derived the dual problem and showed: $\text{Dual} \leq \text{Primal}$.
- In fact, **strong duality holds**:
$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \mathbf{C}, \pi \rangle = \max_{\mathbf{f}, \mathbf{g}: \mathbf{f} \oplus \mathbf{g} \leq \mathbf{C}} (\mathbf{f}^\top \mathbf{a} + \mathbf{g}^\top \mathbf{b})$$

(follows from linear programming duality + feasibility of OT polytope).
- **Implications:**
 - Guarantees **no duality gap** $\rightarrow$ dual solutions faithfully represent the transport cost.
 - Provides alternative characterization of OT distance (as supremum over potentials).
 - Dual variables $\mathbf{f}, \mathbf{g}$ have economic interpretation: “prices” assigned to supply and demand.
- Bridge to continuous case:
 - Sets up the language of c -conjugacy and c -convexity (NEXT)
 - Leads to Kantorovich–Rubinstein duality (continuous analog).
- **Takeaway:** Duality is not just a computational trick — it’s a **structural lens** to understand OT

Section 3: Entropy Regularization